MANAGERIAL ECONOMICS

Decisions Making Under Uncertainty-Assessing Risk

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Decisions Making Under Uncertainty

- Life is full of uncertainty. Will you get a good summer job? Will you be in a car accident? Will the shares of stock you've bought increase in value? Will you avoid earthquakes and floods?

- Managers in all types of business confront a wide variety of risks, including the danger of fire and theft, exchange rate fluctuations, economic downturns, and adverse scientific findings about their products.

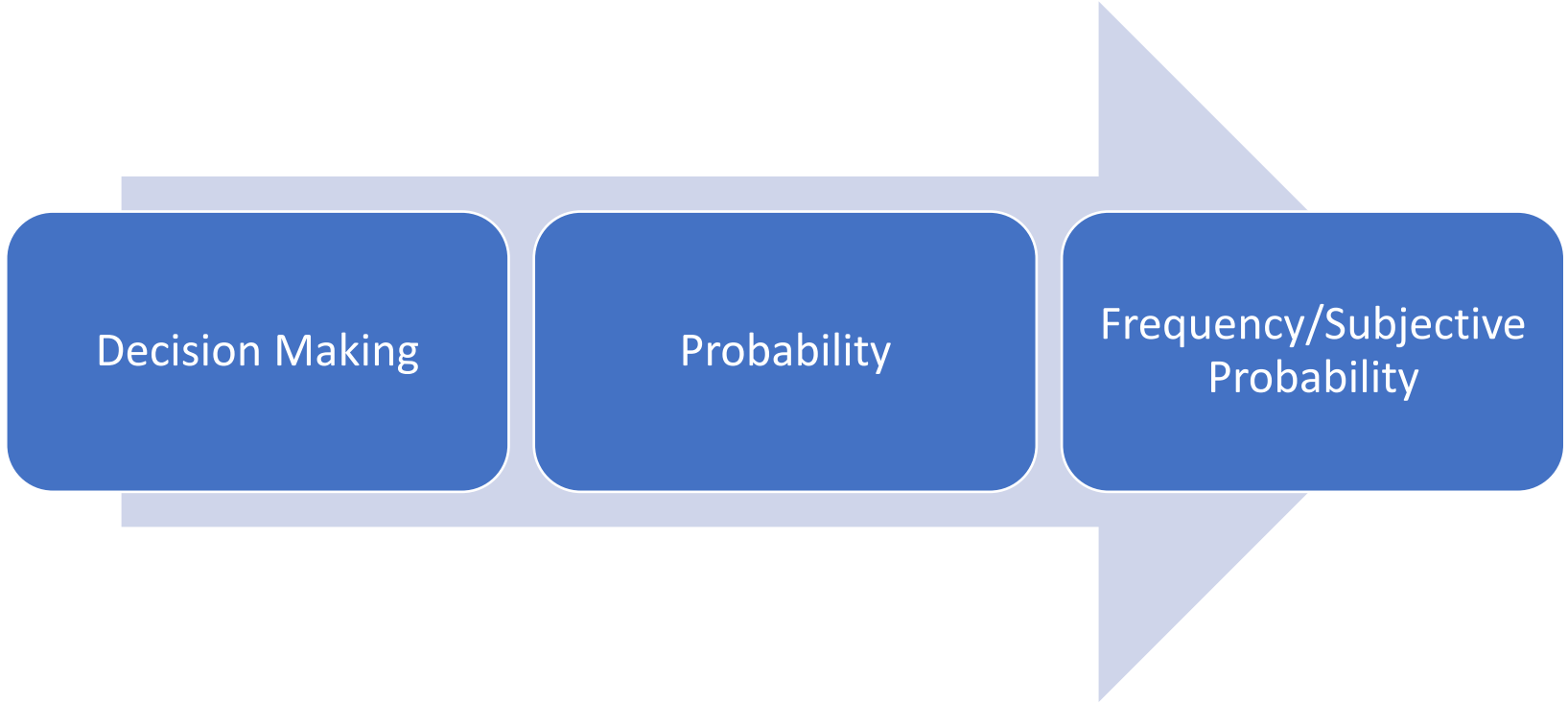
- Some managers dread having to make risky decisions, because if the outcome is bad, they may be fired or have their pay cut.

Decisions Making Under Uncertainty

- Often, we can **quantify an uncertain situation**— one in which no single outcome is certain to occur—because we can assign a probability to each possible outcome.
- For example, if we toss a coin, we can assign a probability of 50% to each of the two possible outcomes: heads or tails. Some people refer to quantifiable uncertainty as risk.
- Consumers and firms modify their **decisions about consumption and investment** as the degree of risk varies. Indeed, most people are willing to spend money to reduce risk by buying insurance or taking preventive measures. Moreover, most people will choose a riskier investment over a less risky one only if they expect a higher return from the riskier investment.

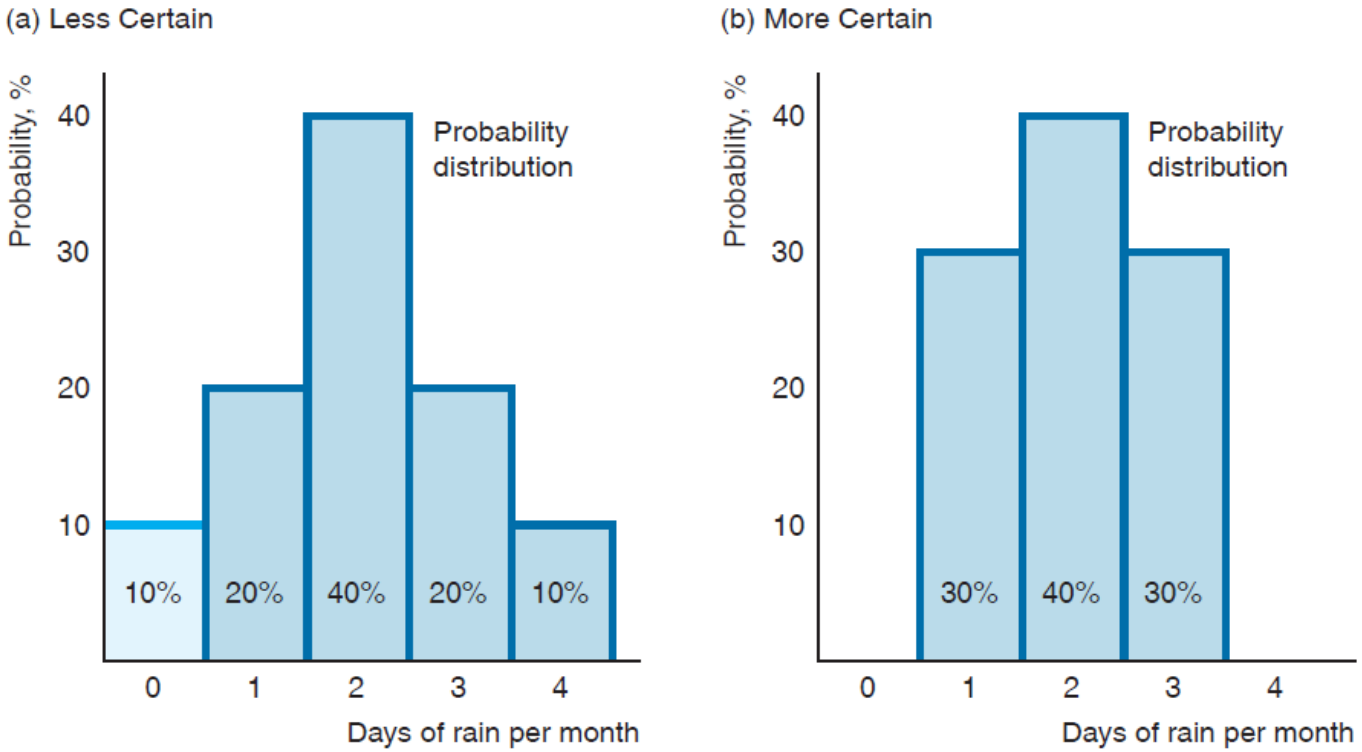
Assessing Risk

- Gregg, a promoter, is considering whether to schedule an outdoor concert on July 4th. Booking the concert is a gamble: He stands to make a tidy profit if the weather is good, but he'll lose a substantial amount if it rains.



Probability

- **Mutually exclusive**-Only one of these outcomes can occur: Either it rains or it does not rain.
- ***Exhaustive***- no other outcomes are possible.
- **Probability Distributions-** A probability distribution relates the probability of occurrence to each possible outcome.



Expected Value

Gregg's earnings from his outdoor concert will depend on the weather. If it doesn't rain, his profit or value from the concert is V = 15 ($15,000). If it rains, he'll have to cancel the concert and he will lose the money, V = -5 ($5,000), that he must pay the band. Although Gregg does not know with certainty what the weather will be, he knows that the weather department forecasts a 50% chance of rain.

TABLE 14.1 Variance and Standard Deviation: Measures of Risk

Outcome	Probability	Value	Deviation = Value − 5	Deviation^2	Deviation^2 × Probability
No rain	$\frac{1}{2}$	15	10	100	50
Rain	$\frac{1}{2}$	−5	−10	100	50
				Variance	100
				Standard Deviation	10

Note: Deviation = Value − EV.

Mini-Case

Risk of a Cyberattack

Cyberattacks—attempts to gain illegal access to computer systems to steal information or harm the system—are one of the newest and largest sources of risk facing major corporations. A cyberattack on Target Corporation exposed personal information of nearly 70 million of their customers. News of the attack resulted in reduced customer traffic and many expenses. Target's earnings before interest and taxes fell by nearly 30%, or $1.58 billion, from the year before the attack. Its breach-related expenses were $292 million, including the settlement of class action lawsuits. After the 2017 announcement of the cyberattack on Equifax, a consumer credit report firm, its stock prices fell by almost one-quarter.

Which firms should put a high probability on an attack? According to Kamiya et al. (2018), cyberattacks are more likely to afflict large, visible firms; highly valued firms; firms with more intangible assets; and firms whose board pays inadequate attention to risk management. They also found that firms suffer major losses when consumer financial information is stolen, but attacks have relatively little effect otherwise.

Attitudes Toward Risk

- Given the risks Gregg faces if he schedules a concert, will Gregg stage the concert? To answer this question, we need to know Gregg's attitude toward risk.
- Expected Utility-Gregg cares about risk as well as expected value. Indeed, most people are risk averse—they dislike risk. They will choose a riskier option over a less risky option only if the expected value of the riskier option is sufficiently higher than that of the less risky one.

$$
\begin{aligned}
EU &= [Pr(\text{no rain}) \times U(\text{Value}(\text{no rain}))] + [Pr(\text{rain}) \times U(\text{Value}(\text{rain}))] \\
&= \left[\tfrac{1}{2} \times U(15)\right] + \left[\tfrac{1}{2} \times U(-5)\right],
\end{aligned}
$$

Attitudes Toward Risk

If we know how an individual's utility increases with wealth, we can determine how that person reacts to risky situations. We refer to a risky situation as a *bet*. Thus, for example, if Gregg schedules his concert outdoors, he is betting that it will not rain. We can classify people based on their willingness to make a **fair bet**: a bet with an expected value of zero. An example of a fair bet is one in which you pay a dollar if a flipped coin comes up heads and receive a dollar if it comes up tails. Because you expect to win half the time and lose half the time, the expected value of this bet is zero:

$$\left[\tfrac{1}{2} \times (-1)\right] + \left[\tfrac{1}{2} \times 1\right] = 0.$$

In contrast, a bet in which you pay \$2 if you lose the coin flip and receive \$4 if you win is an unfair bet that favors you, with an expected value of

$$\left[\tfrac{1}{2} \times (-2)\right] + \left[\tfrac{1}{2} \times 4\right] = 1.$$

Someone who is unwilling to make a fair bet is **risk averse**. A person who is indifferent about making a fair bet is **risk neutral**. A person who is **risk preferring** is always willing to make a fair bet.[8]

Attitudes Toward Risk-Risk Aversion

- We can use our expected utility model to examine how Irma, who is risk averse, makes a choice under uncertainty.
- Figure 14.2 shows Irma's utility function.
- **The utility function is concave** to the wealth axis, indicating that Irma's ***utility rises with wealth but at a diminishing rate***.
- She has diminishing marginal utility of wealth: The extra pleasure from each extra dollar of wealth is smaller than the extra pleasure from the previous dollar.
- An individual whose utility function is concave to the wealth axis is **risk averse**.

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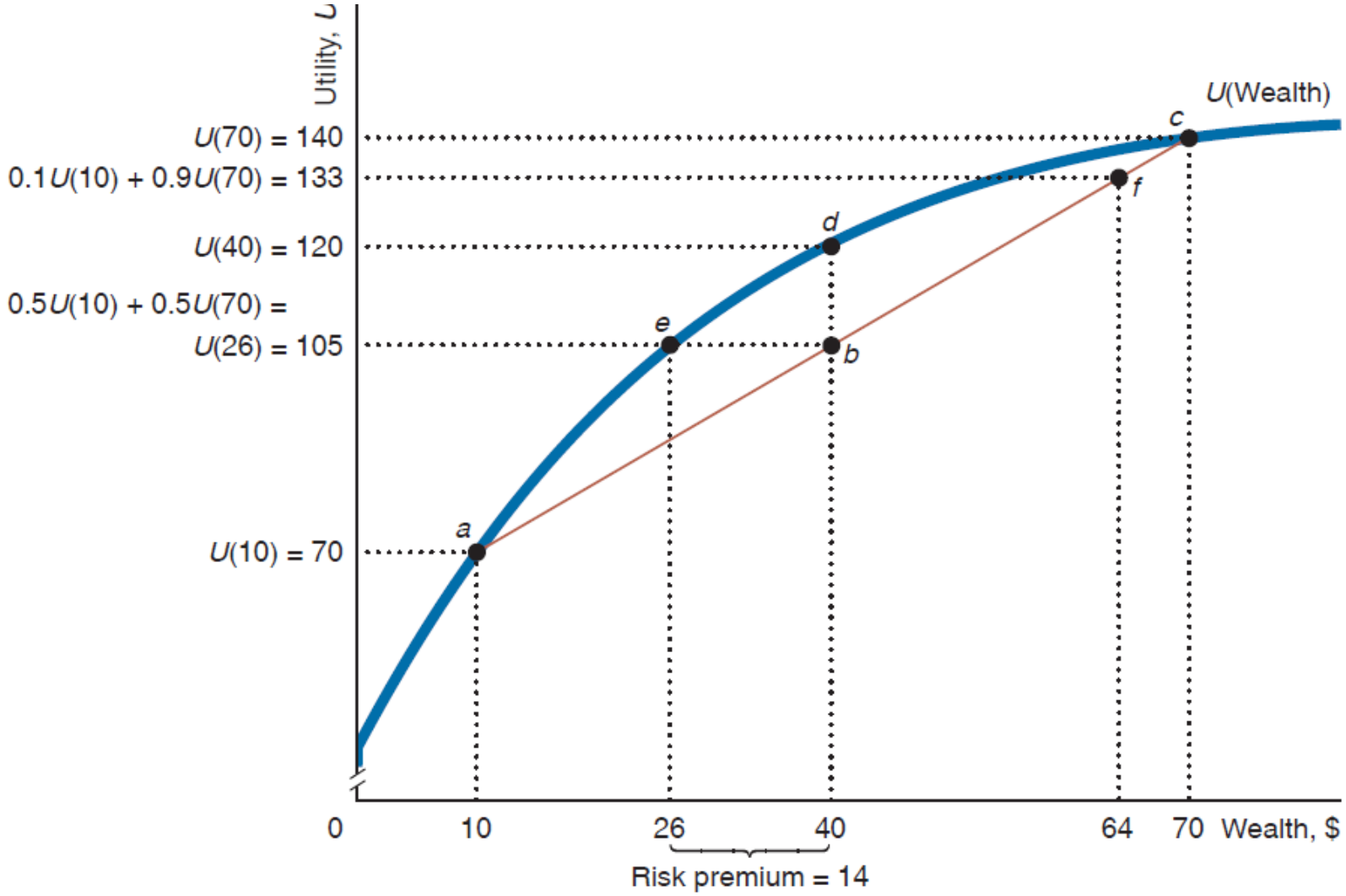
Utility Function and Risk Aversion



Irma's subjective utility is U(40)=120. Would Irma take the risk of buying a stock worth $10 that can increase its value to $70 with 50% probability?

Attitudes Toward Risk

- Initially, Irma's wealth is 40, so her utility is U(40) = 120, at point d.
- If she buys the stock and it's worth 70, her utility is U(70) = 140 at point c. If she buys the stock and it's worth only 10, she is at point a, where U(10) = 70. If her subjective probability that the stock will be worth 70 is 50%, her **expected value** of the stock is 40 = (0.5 * 10) + (0.5 * 70).
- Her **expected utility** from buying the stock is 0.5U(10) + 0.5U(70) = 105, at point b, which is the midpoint of the line between the good outcome, point c, and the bad outcome, point a. Thus, her expected utility from buying the stock, 105, is less than her utility from having a certain wealth of 40, U(40) = 120, so she does not buy the stock.
- In contrast, if Irma's subjective probability that the stock will be worth 70 is 90%, her expected utility from buying the stock is 0.1U(10) + 0.9U(70) = 133, point f, which is more than her utility with a certain wealth of 40, U(40) = 120, at d, so she buys the stock.

Risk Aversion

We can use our expected utility model to examine how Irma, who is risk averse, makes a choice under uncertainty. Figure 14.2 shows Irma's utility function.

- **Unwillingness to Take a Fair Bet.**

A person with a concave utility function would be unwilling to take a fair bet.

- **The Risk Premium.**

The **risk premium** is the maximum amount that a decision maker would pay to avoid taking a risk. Equivalently, the risk premium is the minimum extra compensation (premium) that a decision maker would require to incur a risk.

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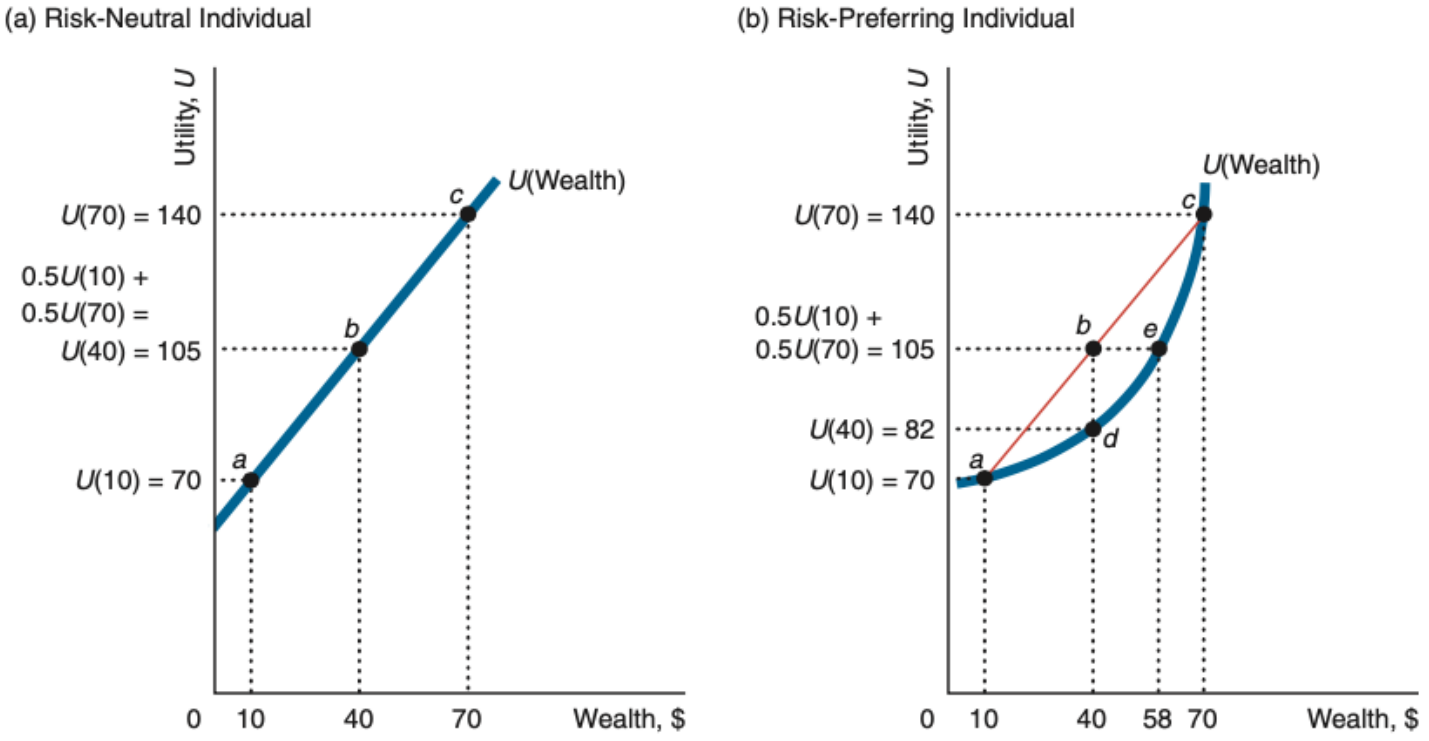
Attitudes Toward Risk

FIGURE 14.3 Risk Neutrality and Risk Preference

(a) If the plot of Irma's utility function is a straight line, she is risk neutral and is indifferent as to whether or not to make a fair bet. Her expected utility from buying the stock, 105 at *b*, is the same as from a certain wealth of 40 at *b*.

(b) If the plot of Irma's utility function is convex to the horizontal axis, Irma has increasing marginal utility of wealth and is risk preferring. She buys the stock because her expected utility from buying the stock, 105 at *b*, is higher than her utility from a certain wealth of 40, 82 at *d*.



Attitudes Toward Risk

Mini-Case

Stocks' Risk Premium

The value of most stocks is more variable over time than that of bonds. Because stocks are riskier than bonds, for both to sell in the market to risk-averse investors, the anticipated rates of return on investing in stocks must exceed those on bonds over the period that the investor plans to hold these investments. This greater return is an investor's risk premium for stocks.

For example, a U.S. government bond is essentially free of any risk that the U.S. government will default. As Figure 14.2 illustrates, an investor will buy a stock only if it provides a risk premium over a risk-free U.S. government bond. That is, the investor buys the stock only if the expected return on the stock exceeds the rate of return on the bond.

In 2017, the stocks in the Standard and Poor's index of 500 leading stocks, the S&P 500, had a return of 21.6%, which exceeded the 2.8% return on 10-year U.S. Treasury bonds by a large margin. However, stocks do not always outperform safe government bonds. In certain years, such as 2008 and 2011, stocks have performed worse than bonds.

Nonetheless, stocks have had a higher rate of return over the long run. For the 50-year period 1968–2017, the annualized return was 10.0% for S&P 500 stocks and 6.8% on long-term bonds.

ATTITUDES TOWARD RISK

Risk Attitudes of Managers

Some shareholders want managers to be risk neutral and to maximize expected profit. However, for a variety of reasons, managers may make either risk-averse or risk-preferring decisions even if shareholders prefer risk-neutral decisions.

It is also possible that shareholders might want managers to behave in a risk- averse manner.

THANK YOU

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